

Aligning a Frozen Transformer with an External Memory

Comparison with Naive Finetuning Baselines

Experiment report — Memory Experiment Sweep (v2 evaluation)

Abstract

We study inference-time alignment of a pretrained 12-layer GPT-NeoX on decision chains, where a coin flip picks one of two decision functions (f or g) at each step. An external key/value memory module ($\sim 295\text{K}$ trainable parameters) is hooked into an early residual-stream layer of the frozen 30M-parameter base model. Evaluated on a dedicated 2,000-example disjoint ffff validation set and 2,000 mixed-pattern examples, the memory achieves 97% f-selection (+45 pp over baseline) and 88% full alignment (+79 pp), with only -1.5 pp arithmetic regression. A single memory entry ($N=1$, 263K params) reaches 95% f-selection while *improving* arithmetic by $+1.5$ pp. Among finetuning baselines at matched compute, LoRA B comes closest but the memory preserves arithmetic $4\times$ better.

1 Problem Setup

Base task. Each example is a decision chain: from a vector $v_0 \in \mathbb{Z}_{10}^6$, the model generates 3–5 blocks (ℓ_k, b_k, v_k) where ℓ_k is a letter naming one of 20 transformations, b_k is step-by-step arithmetic, and v_k is the result. At each step, one of two decision functions selects the next letter:

$$f(v) = \begin{cases} 10 + v_1 & \text{if } v_0 \text{ is even,} \\ v_1 & \text{otherwise,} \end{cases} \quad g(v) = \begin{cases} 10 + v_4 & \text{if } v_3 \text{ is even,} \\ v_4 & \text{otherwise.} \end{cases}$$

Both map a vector to an index in $\{0, \dots, 19\}$, selecting one of 20 letter-ops. During pretraining, a fair coin flip chooses f or g at each step.

Alignment goal. Always pick f 's letter. Base model frozen.

Evaluation. A dedicated validation set of 1,998 ffff-only examples was generated with letter-tuple and input-vector disjointness from training verified (0 overlap). A second mixed-pattern set of 2,000 examples covers all coin-flip combinations. All numbers below are from these v2 evaluation sets.

Metrics

We report three metrics, each measuring a different aspect of alignment quality. All are computed by greedy autoregressive generation followed by scoring against the known decision functions.

1. **f-selection (%)** — the fraction of individual decision-point steps (across all examples) where the model's chosen letter matches $f(v_{k-1})$. This is a *per-step average*: a 4-step chain contributes 4 data points. The pretrained baseline sits at $\sim 52.5\%$ on mixed val (roughly a coin flip per step) and $\sim 77\%$ on ffff val (the model was pretrained on traces that include f-paths, so it partially favours f on inputs whose GT happened to be all- f).
2. **Full alignment (%)** — the fraction of *examples* where **every** step picked f 's letter. This is the strict per-example metric: a chain with 4 out of 5 correct steps scores 0. Because it is a conjunction over steps, the baseline is much lower than f-selection: at $\sim 52.5\%$ per-step probability, $P(\text{all-}f) = 0.525^n$, giving $\sim 14.5\%$ for 3-step chains, $\sim 7.6\%$ for 4-step, and $\sim 4.0\%$ for 5-step. The observed baseline is 8.8% on mixed val, matching this calculation. This metric is the most demanding test of alignment.
3. **Operation accuracy (%)** — the fraction of generated blocks where the arithmetic is correct. For each block, we re-execute the model's chosen letter's transformation on the current intermediate vector and compare the full block text (all arithmetic steps and the result vector). This metric must *not regress* meaningfully — an alignment method that breaks arithmetic is useless. The pretrained baseline sits at $\sim 95\%$.

Two evaluation slices. We report on both:

- **Full val** (2,000 mixed-pattern examples) — the honest alignment test. The model encounters hidden states from all coin-flip histories, including trajectories the memory may not have seen during training.
- **ffff val** (1,998 all- f examples, disjoint from training) — the in-distribution test where `complete_solution` is interpretable (the GT output vector matches the f -path answer).

2 Methods

Memory module. External attention head with $K, V \in \mathbb{R}^{N \times d}$, $W_q \in \mathbb{R}^{d \times d}$, gate $\gamma \in \mathbb{R}$:

$$h' = h + \gamma \cdot \text{softmax}(W_q h \cdot K^\top / \sqrt{d}) V$$

Hooked at layer L of the frozen model. Trained with full-sequence CE on 3,000 mixed traces with f -override at decision points. AdamW, lr 3×10^{-3} , batch size 4, 10 epochs.

FT baselines. Full finetuning (37.9M params) and LoRA rank-8 (295K params) on attention projections. Mode A: ffff-only data with GT labels. Mode B: all data with f -override (same supervision as memory). Note: FT numbers are from the earlier 200-example evaluation; v2 re-evaluation is in progress.

3 Main Results

Table 1: **Alignment comparison on full val (2,000 mixed examples).** Multi-seed entries show mean $\pm$ std. Subscripts: lift over baseline in pp. Baseline: f-selection 52.5%, full alignment 8.8%, op. accuracy 94.8%.

Method	Params	Time	f-selection (%)	Full align. (%)	Op. acc. (%)
Pretrained baseline	–	–	52.5	8.8	94.8
Memory L=1, N=32 (3s)	294 K	177 s	97.2$\pm$0.2 _{+44.8}	88.0$\pm$1.1 _{+79.1}	93.8$\pm$0.9 _{−1.0}
Memory L=2, N=32 (3s)	294 K	220 s	97.4 $\pm$ 0.3 _{+44.9}	88.0 $\pm$ 1.3 _{+79.2}	93.3 $\pm$ 0.1 _{−1.5}
Memory L=0, N=32 (3s)	294 K	244 s	96.1 $\pm$ 0.5 _{+43.6}	82.4 $\pm$ 1.7 _{+73.6}	91.8 $\pm$ 0.9 _{−3.0}
Memory L=2, N=1 (3s)	263 K	211 s	94.8 $\pm$ 0.1 _{+42.3}	76.5 $\pm$ 0.6 _{+67.7}	96.3$\pm$0.4 _{+1.5}
Memory L=2, N=16, n=2000	278 K	111 s	97.8 _{+45.3}	89.0 _{+80.2}	93.3 _{−1.5}
FT full, mode A [†]	37 900 K	335 s	94.8 _{+42.3}	79.0 _{+71.0}	91.4 _{−3.9}
FT full, mode B [†]	37 900 K	355 s	92.5 _{+40.0}	71.5 _{+63.5}	84.3 _{−11.0}
FT LoRA, mode A [†]	295 K	297 s	80.9 _{+28.4}	34.0 _{+26.0}	94.5 _{−0.8}
FT LoRA, mode B [†]	295 K	302 s	96.4 _{+43.9}	83.5 _{+75.5}	89.6 _{−5.7}

[†]FT numbers from earlier 200-example eval; v2 re-evaluation (2k val) in progress.

4 Layer Ablation

5 Memory Size $\times$ Layer Grid

6 Operation Accuracy

7 Per-Chain-Length Analysis

The model generates variable-length traces. For chains where the model produced the expected 3–5 blocks (matching GT length), alignment is near-perfect ($> 99\%$ f-selection, $> 98\%$ full alignment). When the model overgenerated (6+ blocks beyond the GT length of 3–5), f-selection degrades gradually: $\sim 99\%$ at 6 blocks, $\sim 96\%$

Table 2: **Injection layer sweep** at $N=32$, $n=3000$, $e=10$. Earlier layers provide more alignment with a modest op. accuracy trade-off. $L=1$ and $L=2$ are statistically tied on alignment; $L=1$ has the smallest op. accuracy cost.

Layer	Δ f-sel (pp)	Δ full align (pp)	Δ op. acc (pp)	Eval
0	$+43.6 \pm 0.5$	$+73.6 \pm 1.7$	-3.0 ± 0.9	2k, 3 seeds
1	$+44.8 \pm 0.2$	$+79.1 \pm 1.1$	-1.0 ± 0.9	2k, 3 seeds
2	$+44.9 \pm 0.3$	$+79.2 \pm 1.3$	-1.5 ± 0.1	2k, 3 seeds
4	+38.7	+58.0	-1.1	50, 1 seed
6	+36.7	+52.0	-5.2	50, 1 seed
8	+26.1	+28.0	-1.9	50, 1 seed
10	+17.5	+6.0	-0.5	50, 1 seed
11	+20.5	+16.0	-4.8	50, 1 seed

Table 3: **Full alignment (%)** across layers 0–2 and memory sizes $N \in \{1, \dots, 64\}$ at $n = 3000$. All single-seed, $n_{\text{eval}} = 300$. Baseline: 8%. Bold = row best.

Layer	$N=1$	$N=4$	$N=8$	$N=16$	$N=32$	$N=64$
0	61	83	76	79	82	79
1	73	88	86	88	91	80
2	74	89	90	87	91	90

at 8, $\sim 91\%$ at 9. Full alignment drops more sharply (98% at 6, 67% at 8, 41% at 9) because a single wrong pick in a longer chain breaks the strict metric.

Operation accuracy stays at 100% for generated lengths 3–5, starts dropping at 6+ from cascading arithmetic on off-path vectors.

Key observation: the memory generalizes to chains longer than training data (which only contains 3–5 steps), though alignment degrades with length as expected from error cascading.

8 Discussion

Layer 1 $\approx$ Layer 2 $>$ Layer 0 $\gg$ Layer 10. Layers 1 and 2 are statistically indistinguishable on alignment (+44.8 vs. +44.9 pp f-selection). Layer 1 has a marginal edge on op. accuracy (-0.7 vs. -1.5 pp), though within noise (± 1.1 std). Layer 0 is ~ 1.3 pp behind on f-selection and pays a higher arithmetic cost (-3.0 pp). The observation from earlier ablations that “earlier is better” refines to: layers 1–2 are optimal, not layer 0.

$N=1$ as a diagnostic. A single memory entry (one key-value pair, 263K params) reaches 95% f-selection with improved arithmetic. This implies the pretrained model’s f/g preference is approximately controlled by a single direction in layer-2 residual-stream space — consistent with the linear representation hypothesis for model internals on simple tasks.

Memory beats LoRA B at matched params. Both have ~ 295 K trainable parameters and use the same f -override supervision. Memory leads by +1.0 pp f-selection, +3.7 pp full alignment, and +4.2 pp op. accuracy. The gap is architectural: single-attention-head injection at an early layer vs. distributed low-rank deltas across all blocks.

Mode B hurts full FT. With 37.9M free parameters, the f -override on mixed data breaks arithmetic (-11 pp). The model sees a token that disagrees with its own emitted context. With ~ 295 K parameters (LoRA or memory), the same signal is a focused update.

Limitations. Synthetic task, small model (30M). FT baselines await re-evaluation on the v2 validation sets. Chain-length breakdown uses generated (not GT) lengths; GT lengths are 3–5 only.

Table 4: **Arithmetic preservation** on full val (2k examples). Memory N=1 is the only aligning method that *improves* arithmetic. FT full mode B suffers catastrophic forgetting (−11 pp).

Method	Baseline	Trained	Δ
Memory L=2, N=1 (3s)	94.8 %	96.3 %	+1.5 pp
Memory L=1, N=32 (2s)	94.8 %	94.1 %	−0.7 pp
Memory L=2, N=32 (3s)	94.8 %	93.3 %	−1.5 pp
Memory L=0, N=32 (3s)	94.8 %	91.8 %	−3.0 pp
FT LoRA, mode A [†]	95.3 %	94.5 %	−0.8 pp
FT LoRA, mode B [†]	95.3 %	89.6 %	−5.7 pp
FT full, mode A [†]	95.3 %	91.4 %	−3.9 pp
FT full, mode B [†]	95.3 %	84.3 %	−11.0 pp

[†]FT baselines from 200-example eval.

9 Conclusion

An external memory module (294K params, 3 min training) matches or beats full-model finetuning (37.9M params) on alignment, preserves arithmetic better, and is fully reversible. The critical architectural choice is injecting at layers 1–2 (+45 pp alignment) rather than later layers. A single memory entry (263K params) already achieves 95% f-selection while improving arithmetic accuracy.